

A Controlled Semantic-Intervention Protocol for Construct Validation of Ontology-Alignment Justification Metrics

Vitaliy Charushin¹

¹*Independent Researcher, Paris, France*

Abstract

Evaluation of ontology alignment is dominated by correspondence correctness — precision, recall, and F-measure against a reference alignment. As large language models increasingly produce not only correspondences but natural-language *justifications* for them, a complementary question arises: does a proposed metric of justification *quality* actually measure what it claims? We contribute a controlled semantic-intervention protocol for the *construct validation* of coherence-oriented justification metrics, and instantiate it on a frozen benchmark of 20 candidate correspondences from three real OAEI Conference ontology pairs (SCI-OM20). Each correspondence receives a justification subjected to five component-targeted semantic interventions, each paired with a meaning-preserving control at a matched locus, yielding 220 blinded artifacts scored under a frozen rubric. Using a five-dimensional coherence instrument (the Semantic Coherence Index, SCI) as the first metric validated with the protocol, we find that every intervention reliably decreases its designated dimension (H1); selectivity is structured but partial — four of five interventions are diagonally dominant, with interpretable cross-dimensional coupling (H2); aggregate coherence declines under perturbation despite fixed correctness labels, ruling out simple re-encoding of correctness (H3); and, conditional on intervention type, edit magnitude is unrelated to the response (H4). A second, independently run evaluator produces strongly concordant scores (quadratic-weighted $\kappa = 0.73$; 96% within one ordinal anchor), reducing the likelihood that the structure is idiosyncratic to a single judge. The protocol is metric-agnostic and applies to other justification-quality instruments.

Keywords

ontology matching, alignment justification, construct validity, semantic intervention, evaluation methodology, LLM-as-judge

1. Introduction

Ontology matching produces correspondences between the entities of two ontologies, and its evaluation is organized almost entirely around *correctness*: whether a proposed correspondence appears in a curated reference alignment, summarized by precision, recall, and F-measure [1]. This is appropriate for the correspondence itself, but increasingly incomplete: large language models now participate throughout the matching pipeline and produce, alongside each correspondence, a natural-language *justification* citing labels, definitions, axioms, and hierarchical context. A correspondence can be correct while its justification is incoherent, and incorrect while its justification is superficially plausible. Justification quality is therefore a distinct evaluation target, but methods for validating instruments intended to measure it remain underdeveloped.

Before any such measure is deployed, a prior question must be answered: does it *measure what it claims*? In the measurement sciences this is the problem of *construct validity* [2, 3]: construct validation requires convergent and discriminant evidence that scores vary with the intended construct rather than with irrelevant factors. We operationalize this requirement through controlled semantic interventions paired with matched controls.

Common validation strategies — face validity, qualitative examples, or correlation with human judgments — can demonstrate usefulness but do not establish that an instrument responds *selectively* to the semantic property it claims to measure; our protocol addresses this gap through controlled semantic interventions.

We contribute a **controlled semantic-intervention protocol** for construct-validating justification-quality metrics. Given a correspondence and its justification, the protocol applies a taxonomy of

OM 2026: The 21st International Workshop on Ontology Matching collocated with the 25th International Semantic Web Conference (ISWC 2026), October 25th, 2026, Bari, Italy

 0000-0002-5119-7515 (V. Charushin)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

component-targeted semantic interventions — each damaging one hypothesized dimension of justification quality while holding the correspondence and its conclusion fixed — and pairs each intervention with a meaning-preserving control at a matched textual locus. A valid instrument should respond to each intervention on its designated dimension, should not respond to the matched controls, and should track the *semantic function* of an edit rather than its surface magnitude. The protocol is agnostic to the metric under test; the same interventions apply to any future coherence metric, symbolic evaluator, or human rubric.

We instantiate the protocol with a five-dimensional coherence instrument, the Semantic Coherence Index (SCI), as the first metric validated under it, on a frozen benchmark (SCI-OM20) of 20 candidate correspondences from the OAEI Conference track. The contributions are: (i) the intervention protocol and its matched-control design; (ii) a frozen, reproducible benchmark of 220 blinded artifacts with a prospectively frozen analysis plan; and (iii) a construct-validity study whose scoring robustness is assessed with a second evaluator run independently. The study shows that the instrument exhibits structured, partially selective sensitivity to semantic damage and captures a dimension of justification quality not reducible to correspondence correctness. We do *not* evaluate matcher robustness: our artifacts are controlled justifications, not operational outputs; extending the protocol to real matcher outputs is the natural sequel (Section 8).

2. Related Work

LLMs in ontology matching. A growing body of work applies LLMs to matching through prompting, retrieval, structural context, and agent-based pipelines [4, 5, 14], evaluated mostly on correspondence performance, sometimes with hallucination analysis [6]. That line asks whether generated correspondences are *correct or hallucinated*; we ask the orthogonal question of whether a *metric of justification quality* behaves as a valid instrument under controlled semantic damage.

Alignment coherence and explanation. The community has studied the logical coherence of alignments and their repair [8, 9], and the importance of explanation and provenance for human validation, though many systems still expose only confidence values [10]. Our notion of coherence is not the logical consistency of a mapping set but the *structural quality of the justification* for a single correspondence. Argument quality has rich taxonomies in natural-language argumentation [7]; our object is a single alignment’s justification, and we validate the measuring instrument *experimentally* rather than only correlating it with human ratings.

Construct validity. The evaluation logic we adopt is psychometric: an instrument is construct-valid to the extent that it responds to targeted manipulations of the intended construct and not to irrelevant variation [2, 3]. Controlled perturbations and contrast sets test whether NLP systems exhibit intended behavioral capabilities [13]; we apply related logic to construct-validating an evaluation instrument rather than a model’s accuracy. LLM-as-judge methods offer scalable rubric-based evaluation but are known to exhibit evaluator-specific biases [12], motivating our cross-evaluator robustness check. Applying this discipline — frozen protocol, pre-specified predictions, matched controls, blinded scoring — to an LLM-scored metric is, to our knowledge, novel in ontology matching.

3. Methodology

3.1. The evaluated object and the instrument

The protocol evaluates the pair (M, J) of a candidate correspondence M and its justification J , conditioned on the source and target ontologies and the governing context C : $\text{SCI}(M, J \mid O_s, O_t, C)$, where C denotes the governing evaluation context, such as the relevant ontology sub-vocabulary or scope under which the correspondence is interpreted. Across all variants of a case, M and its reference-alignment correctness label are held fixed; only J is edited. The instrument under test scores J on five interpretable dimensions, each in $[0, 1]$: **TR** (Ontology Traceability), **EA** (Evidence Alignment), **LC** (Level Consistency), **CS** (Counterfactual Sanity), and **CDS** (Cross-Context Stability).

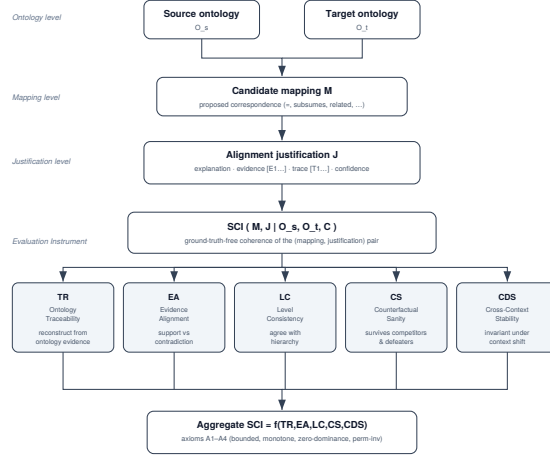


Figure 1: The evaluated object and instrument. Source/target ontologies yield a candidate correspondence M and its justification J ; the instrument scores (M, J) on five dimensions (TR, EA, LC, CS, CDS) that combine into an aggregate. Interventions damage one dimension of J while M and the conclusion are held fixed.

Component scores use a discrete five-anchor scale $\{0, 0.25, 0.5, 0.75, 1\}$. For hypotheses requiring a composite, we use the frozen geometric-mean aggregate $SCI = (\prod_k s_k)^{1/5}$, which is bounded, monotone, permutation-invariant, and zero-dominant; per-dimension analyses do not depend on the aggregate.

The five dimensions were chosen to represent distinct failure loci in an alignment justification: whether the mapping is reconstructible from ontology evidence (TR), whether that evidence supports rather than contradicts the claim (EA), whether the claim is compatible with the relevant hierarchy (LC), whether it survives plausible competitors or defeaters (CS), and whether it remains stable under legitimate changes of context (CDS). We do not claim this decomposition is exhaustive or uniquely optimal; the study tests only whether these hypothesized dimensions produce distinguishable response patterns under intervention.

3.2. Benchmark: SCI-OM20

We draw 20 candidate correspondences between entities from three ontology pairs of the OAEI Conference track (OntoFarm) [11]: `cmt-ekaw`, `cmt-sigkdd`, and `ekaw-sigkdd`. Correctness labels are frozen against the `ra1` reference alignment, yielding a balanced 10/10 split of reference-positive and reference-negative correspondences. Negatives are surface-plausible but reference-rejected (e.g., `cmt:Acceptance` vs. `ekaw:Accepted_Paper` — a decision versus a document). Ontology content used in an artifact (identifiers, superclasses, axioms, comments) is transcribed verbatim. Each correspondence receives a sincere baseline justification; for negatives the flaw is latent in the axioms, not annotated, preserving the blind.

3.3. Intervention taxonomy

Each baseline is subjected to five component-targeted interventions, all retaining the conclusion verbatim (Fig. 1):

- **TR (obscure):** replace the justification with same-length wording that cites no reconstructible ontology evidence.
- **EA (contradict):** add an evidence item contradicting a key premise (a disjointness/type/extension conflict).
- **LC (level clash):** replace a reasoning step so that it exposes a parent/hierarchy inconsistency.
- **CS (defeater):** introduce an unaddressed discriminating competitor or type defeater.
- **CDS (context shift):** restrict or expand the governing scope using only real sub-vocabulary, so applicability is placed in question without contradicting any fixed axiom.

3.4. Matched controls

Every intervention is paired with a meaning-preserving control at the same locus: a paraphrase of the removed content (cTR), a neutral non-probative addition (cEA), a paraphrase of the altered step (cLC), a non-discriminating alternative (cCS), and a scope restatement that does not change applicability (cCDS). Interventions are heterogeneous operations, not equal-size edits; the matched controls establish comparability, and surface distances are reported (Section 5.4). This yields $20 \times (1 + 5 + 5) = 220$ artifacts. The worked example below shows the logic on the running case `cmt:Acceptance` vs. `ekaw:Accepted_Paper` (reference-negative: a decision versus a document).

Variant	Illustration
Baseline	Argues <code>Acceptance = Accepted_Paper</code> from shared labels and review context.
EA intervention	Adds evidence contradicting the equivalence (disjoint types: a decision vs. a document).
cEA control	Adds a neutral ontology fact at the same locus; evidential support unchanged.
Expected response	EA drops sharply under the intervention; stable under cEA.

Intervention and control modify comparable textual material, but only the intervention changes the semantic relation between the evidence and the correspondence. The example is illustrative; the measured component scores come from blinded evaluation of the frozen benchmark artifacts.

4. Experimental Design

4.1. Scoring

Each artifact is scored standalone and blinded: the evaluator sees the governing scope, the source and target entities (labels, superclasses, subclasses, axioms, comments), the proposed relation, and the justification — never the case identity, condition, target dimension, predictions, or matched-artifact identity. Scoring follows the LLM-as-judge paradigm [12], using a rubric with anchors fixed before evaluation, fixed evaluator versions, prompts, and inference configurations, and structured JSON output. The primary evaluator scored all 220 artifacts; because LLM judges can exhibit evaluator-specific biases, a second evaluator, run independently, scored a stratified 66-artifact subset ($\sim 30\%$) to test whether the score structure survives a change of evaluator. The primary evaluator was GPT-5.5; the secondary evaluator was Claude Fable 5. Evaluator identities are recorded for reproducibility only; the analysis tests whether the construct structure survives a change of evaluator, not which model scores better.

4.2. Response measures and pre-specified hypotheses

The primary dependent variable is the per-dimension *control-adjusted delta* $\Delta_{ijk}^c = s_{ijk}^P - s_{ijk}^C$, the difference between a perturbation and its matched control on dimension k for case i and intervention j . Analyses were pre-specified before scoring; results are reported separately for the positive and negative strata. The hypotheses are:

- H1 (targeted sensitivity)** Each intervention produces a negative Δ^c on its designated dimension.
- H2 (selectivity)** The designated dimension shows the largest-magnitude decrease, that is, the most negative mean control-adjusted response; theory-consistent couplings are permitted.
- H3 (non-reducibility to correspondence correctness)** If the instrument merely re-encodes correspondence correctness, semantic perturbations to the justification should not systematically reduce aggregate coherence while the correspondence and its correctness label remain fixed.
- H4 (semantic function over surface magnitude)** SCI responses are determined by the semantic function of an intervention rather than its lexical or surface magnitude.

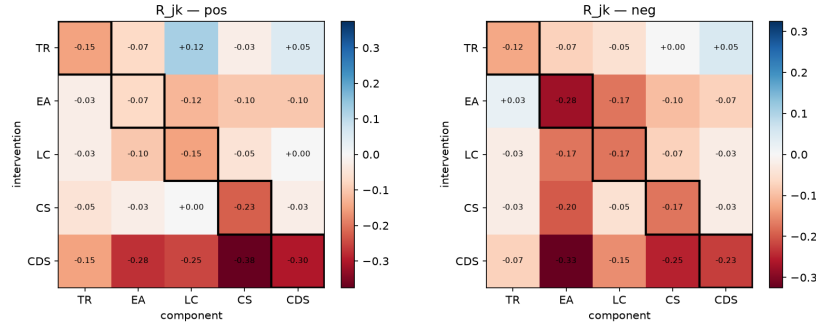


Figure 2: Control-adjusted response matrix R_{jk} (rows: interventions; columns: dimensions), positive and negative strata; a diverging scale is centered at zero, diagonal outlined. On pooled data four of five interventions (TR, EA, LC, CS) are diagonally dominant; CDS acts partly as global damage, most visibly through CS. Per-stratum structure is noisier at $n = 10$ and shows broader off-diagonal effects.

5. Results

5.1. H1: targeted sensitivity

Every intervention reduced its designated dimension, with a 95% bootstrap confidence interval entirely below zero and a significant Wilcoxon signed-rank test ($p < 0.005$); matched-pairs rank-biserial effects are near or at the ceiling (Table 1). Effect estimates range from $\Delta^c = -0.138$ (TR) to -0.263 (CDS).

Table 1

H1: control-adjusted response on the targeted dimension (pooled). Rank-biserial is negative by convention because damage is negative.

Interv.	mean Δ^c	Wilcoxon p	rank-bis.	95% CI
TR	-0.138	0.0020	-1.00	[-0.200, -0.075]
EA	-0.175	0.0010	-1.00	[-0.263, -0.100]
LC	-0.163	0.0049	-0.87	[-0.250, -0.075]
CS	-0.200	0.0005	-1.00	[-0.287, -0.125]
CDS	-0.263	0.0001	-1.00	[-0.338, -0.188]

5.2. H2: selectivity and cross-dimensional coupling

H2 received partial support. After pooling strata, TR, EA, LC, and CS were diagonally dominant, each producing its largest-magnitude decrease on the targeted dimension, whereas CDS was not: contextual perturbation reduced CS (-0.312) more than CDS itself (-0.263). The stratum-specific matrices (Fig. 2) reveal broader cross-dimensional effects: CDS behaves partly as global semantic damage, particularly through CS, while EA in the negative stratum is sensitive to several interventions. The positive TR→LC response in the positive stratum ($+0.12$) shows that a targeted degradation can occasionally raise an off-target score, reinforcing that the dimensions are coupled rather than independent. The response structure is therefore discriminative but not orthogonal.

5.3. H3: non-reducibility to correspondence correctness

With the correspondence and its correctness label fixed by construction, aggregate coherence still declined under perturbation in 71 of 100 case-intervention pairs (36/50 positive, 35/50 negative; mean drops 0.125 and 0.110); fourteen perturbations drove a component to zero, collapsing the aggregate under zero-dominance. Because this occurs in both strata with the reference label fixed, it rules out the hypothesis that the instrument merely re-encodes correctness — though a natural-data dissociation remains future work.

5.4. H4: semantic function over surface magnitude

Matched controls carry the strongest evidence: substantial surface edits in a control produce essentially no response, while perturbations move the targeted dimension sharply — the TR control, at a mean

normalized token-edit distance of 0.49, moves its dimension by only -0.013 , versus -0.150 for the perturbation. Crucially, *conditional on intervention type*, edit magnitude and response are unrelated: within perturbations the Spearman correlation between normalized token-level Levenshtein distance and the absolute target-dimension change ($|s_{ijj}^{\text{cond}} - s_{ijj}^{\text{base}}|$) is $\rho = +0.066$ ($p = 0.52$), within controls $\rho = -0.099$ ($p = 0.33$); a pooled $\rho = +0.22$ reflects only the between-condition edit-size difference. The responses are therefore attributable to semantic function, not lexical magnitude.

5.5. Evaluator robustness

A second evaluator, run independently, scored the frozen stratified subset under the identical protocol. Agreement was substantial across all five dimensions (per-dimension Spearman $\rho = 0.73$ – 0.84 ; quadratic-weighted $\kappa = 0.73$; 96% within one ordinal anchor), indicating the score assignments are largely evaluator-independent, with Level Consistency least reliable (36% exact).

6. Discussion

The study supports H1 and the narrow H3 non-reducibility claim, H4 through the null control responses, and structured but partial selectivity (H2): the dimensions are distinguishable but not orthogonal, the diagonal structure arguing against an undifferentiated error signal while the CDS–CS and EA–LC couplings show attribution is not fully separable. That an independently run second evaluator produces strongly concordant scores reduces the likelihood the structure is idiosyncratic to one judge.

In an operational pipeline, a matcher may emit a correspondence together with a justification; a validated justification metric can then act as a diagnostic layer between matching and acceptance, with low or dimension-specific scores triggering human review, targeted re-reasoning, or rejection of an otherwise high-confidence correspondence. The present study validates the measurement layer; integration with operational matchers is future work.

More broadly, the contribution is the protocol rather than the instrument: it introduces *experimental construct validation* into ontology evaluation, and any future coherence metric or human rubric can be subjected to the same interventions — SCI is simply the first it validates.

7. Threats to Validity

Construct and manipulation strength. Baseline justifications are authored, not emitted by operational matchers, and interventions instantiate clear semantic defects; the study validates the measurement methodology under a strong-form manipulation, so naturally occurring justifications may be subtler or mixed. *Scale.* With twenty cases, one domain, and five anchors, SCI-OM20 is a construct-validation, not a performance, benchmark, so we lead with effect estimates rather than p -values. *Evaluators and human validation.* Both evaluators are closed-source LLMs, and the secondary shares a model family with the artifact-drafting model, so agreement does not establish correspondence with expert human judgment and cannot exclude shared model biases; frozen outputs keep the reported analysis reproducible despite model updates. Human expert scoring and a third-family judge are important next steps. *Baselines.* We do not compare SCI with a one-dimensional holistic justification-quality judge; such a baseline is required to determine whether the multidimensional decomposition provides incremental diagnostic value beyond generic degradation detection. *Aggregation.* H3 uses one admissible (frozen) aggregate; the zero-dominance collapses illustrate the axiom rather than validating it as uniquely correct. *Reproducibility.* The benchmark, intervention/control specifications, evaluator prompt and complete component anchor rubric, hypotheses, analysis plan, randomization order, and both frozen scoring runs are content-hashed and available from the author on request.

8. Conclusion

We presented a metric-agnostic protocol for construct-validating ontology-alignment justification metrics, and showed a five-dimensional coherence instrument that responds with structured, partially selective sensitivity to semantic damage, is not reducible to correctness, and tracks an edit’s semantic function rather than its size. Validation on human judgments, a holistic one-dimensional baseline, and operational matcher outputs remains future work.

Declaration on Generative AI

During the preparation of this work, the author used Anthropic Claude in order to: Drafting content, Abstract drafting, Paraphrase and reword, Improve writing style, Formatting assistance, Citation management, Content enhancement, and Peer review simulation. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the publication's content. The large language models evaluated as scoring instruments in the study are part of the research methodology and are described in the text, not tools used to prepare the manuscript.

References

- [1] J. Euzenat, P. Shvaiko. *Ontology Matching*, 2nd ed. Springer, 2013. doi:10.1007/978-3-642-38721-0.
- [2] L. J. Cronbach, P. E. Meehl. Construct validity in psychological tests. *Psychological Bulletin* 52(4):281–302, 1955. doi:10.1037/h0040957.
- [3] D. T. Campbell, D. W. Fiske. Convergent and discriminant validation by the multitrait–multimethod matrix. *Psychological Bulletin* 56(2):81–105, 1959. doi:10.1037/h0046016.
- [4] H. Babaei Giglou, J. D’Souza, F. Engel, S. Auer. LLMs4OM: Matching Ontologies with Large Language Models. In *The Semantic Web: ESWC 2024 Satellite Events*, LNCS 15344, Springer, pp. 25–35, 2025. doi:10.1007/978-3-031-78952-6_3. arXiv:2404.10317.
- [5] Z. Qiang, W. Wang, K. Taylor. Agent-OM: Leveraging LLM Agents for Ontology Matching. *Proc. VLDB Endowment* 18(3):516–529, 2024. doi:10.14778/3712221.3712222.
- [6] Z. Qiang, K. Taylor, W. Wang, J. Jiang. OAEI-LLM: A Benchmark Dataset for Understanding Large Language Model Hallucinations in Ontology Matching. In *Proc. Special Session on Harmonising Generative AI and Semantic Web Technologies (HGAIS 2024)*, CEUR Workshop Proceedings, vol. 3953, 2025.
- [7] H. Wachsmuth, N. Naderi, Y. Hou, Y. Bilu, V. Prabhakaran, T. Alberdingk Thijm, G. Hirst, B. Stein. Computational Argumentation Quality Assessment in Natural Language. In *Proc. EACL*, pp. 176–187, 2017. doi:10.18653/v1/E17-1017.
- [8] C. Meilicke, H. Stuckenschmidt. Incoherence as a Basis for Measuring the Quality of Ontology Mappings. In *Proc. Ontology Matching (OM-2008)*, CEUR Workshop Proceedings, vol. 431, 2008.
- [9] E. Jiménez-Ruiz, B. Cuenca Grau. LogMap: Logic-Based and Scalable Ontology Matching. In *Proc. ISWC*, LNCS 7031, Springer, pp. 273–288, 2011. doi:10.1007/978-3-642-25073-6_18.
- [10] H. Li, Z. Dragisic, D. Faria, V. Ivanova, E. Jiménez-Ruiz, P. Lambrix, C. Pesquita. User validation in ontology alignment: functional assessment and impact. *The Knowledge Engineering Review* 34:e15, 2019. doi:10.1017/S0269888919000080.
- [11] O. Zamazal, V. Svátek. The Ten-Year OntoFarm and its Fertilization within the Onto-Sphere. *Journal of Web Semantics* 43:46–53, 2017. doi:10.1016/j.websem.2017.01.001.
- [12] L. Zheng et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems* 36, 2023.
- [13] M. T. Ribeiro, T. Wu, C. Guestrin, S. Singh. Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. In *Proc. ACL*, pp. 4902–4912, 2020. doi:10.18653/v1/2020.acl-main.442.
- [14] S. Hertling, H. Paulheim. OLaLa: Ontology Matching with Large Language Models. In *Proc. K-CAP*, pp. 131–139, 2023. doi:10.1145/3587259.3627571.